Plant OS - Project Proposal

Project Title: Plant OS - Verified Plant Identification, Cataloging, and Care Reminders

Team Name: The Practical Coders

Team Members & Roles:

Aran Gain - Project Manager / Front End Developer (UI/UX)

Travis Shepherd - Back End & Integration Developer

Advisor: Prof. Carrie Lapaire Harmon

Advisor Email: clharmon@ufl.edu

Date of Submission: February 1st, 2026

Abstract

A lot of home plant owners (myself included) struggle to keep their plants alive and healthy. Some of the reasons include identifying the plants type incorrectly, having inconsistent routines for caring and watering the plants, and having difficulty finding good advice. Additionally, having differing plants with varying needs of sunlight, moisture, environment, and water makes this task more difficult.

Plant OS is a software application designed to help users correctly identify and keep track of their plants using an external plant identification service, store these plant verified information in a personal catalog in the app, and additionally receive plant-specific care guidance like watering reminders. Plant OS will also recommend credible resources so users can access trustworthy science-based information and local expertise. The system will be built using a modern web technology stack, with an authenticated database, media storage, and external API integration. The semester deliverable will be a functional beta demonstrating the features of verified plant onboarding workflow, plant catalog system, reminder scheduling, and an extension resource section.

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Copy of Team Contract

Team Contract
Team Name: The Practical Coders
Members: Aran Gain, Travis Shepherd
Course: Senior Design, CIS4914

Overall Objectives:
- The objective of our team is to design, test, and deliver a user-focused web and/or mobile application that'll help plant owners accurately identify and care for their plants.
- The system will allow the user to catalog their plants, identify species of plants, and receive guidance for taking care of their plants. For example, it'll send watering reminders specific to each plant, and information that's relevant.
- The goal of our team is to create a fully functional (and documented) demo for a system that'll showcase the application of our skills, and serve a real purpose.

Individual Roles & Responsibilities
- **Team Leads**: Aran Gain & Travis Shepherd
  - Coordinate project planning, milestones, and task assignments
  - Serve as the primary point of contact with the faculty advisor
  - Ensure deliverables align with the course requirements and deadlines
- **Frontend Developer & Conceptual Backend Lead:** Aran Gain
  - Design and implement user interface
  - Develop user flows
  - Ensure accessibility, usability, and consistent visual design
  - Collaborate with backend developer to integrate frontend with backend.
  - Help with backend
- **Backend / Integration Lead:** Travis Shepherd
  - Design and maintain the application backend and database schema
  - Integrate external plant identification APIs
  - Implement authentication, data storage, and logic
  - Ensure performance, reliability, and data integrity
- **Quality Assurance & Documentation:** Aran Gain & Travis Shepherd
  - Develop and execute testing plans
  - Track bugs and issues using GitHub
  - Maintain documentation
  - Create demo & presentation materials

Values and Agreement Statements:
- I agree to communicate clearly and respectfully, and to respond to messages within 24 hours.
- I agree to attend all scheduled meetings unless a reasonable excuse is given.
- I agree to complete assigned tasks by agreed deadlines, and notify the team immediately if there's an issue that can cause delays.
- I agree to maintain a professional and collaborative environment, where feedback is appreciated.

Software Configuration Management Protocol
- The team will use GitHub as the central hub for version control, documentation, and for management.
- Repository Structure: A single primary repository.
- Branching:
  - Main: contains stable, production-ready code
  - Dev: contains integrated features ready for testing
  - Feature/*: created for individual features and/or fixes for bugs.
- Pull Requests: All code must be submitted through a pull request before merging into dev or main.
- Code Review: We must review and approve each other's pull request before merging.
- Merge Conflicts: The developer creating the pull request is responsible for solving/resolving merge conflicts, with the other team member's help.
- Versions: Weekly milestone versions will be tagged.
- Backup & Recovery: The repository will be backed using GitHub and local clones on team members' local machines/computers.

Meeting Times & Days (Team Meetings)
- The team will meet outside of advisor meetings at the following times:
  - Weekly spring planning: Mondays at 2:30pm
  - Progress Review: Wednesdays at 2:30pm
- These times can be adjusted on mutual agreement.

Meetings with Advisor & Facilitator
- The team will aim to meet with the faculty advisor on a biweekly basis on Zoom, although she has suggested limited communication due to busy schedules.
- All team members are required to attend every meeting.

Communication Protocol:
- Primary Communication: Discord
- Communication with Advisor & Facilitator: Email
- Project Tracking: GitHub
- Meetings: Zoom or Google Meets

Conflict Resolution Protocol:
- Team members involved in a conflict will discuss the issue privately to seek a resolution.
- If unresolved, the issue will be brought up to the faculty advisor for guidance and resolution.

Consequences:
- Repeated missed deadlines or meetings will be reflected in peer evaluations.

Signatures
- By signing below, we agree to abide by the terms of this team contract.
- **Aran Gain**
  - Signature: Aran Gain
- **Travis Shepherd**
  - Signature:

1. Introduction and Motivation for the Project

1.1 Problem Formation

Many plant owners experience difficulty with keeping their plants healthy, especially over a long period of time. The reasons include not having correct identification resources for plants, inconsistent care routines for plants, and not having access to trustworthy guidance. If plant care recommendations are made for the wrong species, the guidance becomes unreliable and can worsen the conditions of the plant as well. Even when it is correctly identified, users often follow watering schedules that are generic, which don't reflect the differences between plants and the seasonal conditions.

1.2 Rationale

A practical software tool can reduce these problems through a structured workflow: verify plant identity, store plant information in a personal catalog, and support care habits for plants through reminders (specific to each plant). Plant OS is motivated by the idea that correct identification, good advice, and care reminders will make it easier for plant owners to take care of their plants. The application will integrate an external plant identification service so recommendations are tied to verified species rather than a guessed label, which will then set reminder scheduling to help users maintain consistent habits for plant care.

1.3 Proposed Solution

Plant OS will allow users to:

- Upload a photo of a plant and identify it through an external plant identification service
- Store verified plant identity and details in a personal plant catalog

- Receive plant-specific care guidance and watering reminders
- Access credible extension resources based on region/location

1.4 Importance and Impact

Plant OS aims to help plant owners of all experiences keep their plants healthier through accurate identification and consistent care routines (guidance and reminders). From a SWE perspective, the project provides a realistic scope that includes full-stack development, external API integration, authenticated data storage, and a deployable beta product.

2. Literature Survey (3 Sources)

Source 1: Reminder systems and adherence to repeated routines

This work talks about a behavior model that explains how motivation, ability and prompts help influence user behavior. In our case, we will be using well timed reminders to help users perform care routines for their plants. This supports Plant OS's watering reminders system, which aims to encourage consistent plant care habits.

Source 2: Plant Identification Technologies (computer vision)

This paper surveyed computer vision techniques used for automatic plant species identification. It highlights the effectiveness of convolutional neural networks while talking about the misclassification and reduced accuracy in poor photo conditions. This supports the Plant OS' use of an external plant identification service and the need for user confirmation during the identification process.

Source 3: regionally relevant extension resources

The Congressional Research Service explains that the Cooperative Extension System was created to deliver research-based knowledge from land-grant universities directly to the public through state networks. This work supports Plant OS' choice to link to regionally relevant extension resources within the app. Since plant care recommendations vary significantly by location, connecting users to extension resources based on location and region will be helpful.

3. Proposed Work

3.1 Plan of Work

Each week our goal will be to have an updated, stable build of Plant OS on our github. Our plan is to implement the core functionality first (user authentication, plant onboarding, verified identification, and catalog storage). After this framework is stable, we will integrate the reminder functionality and scheduling, along with a method to access and find local expertise.

3.2 Target Audience

- Home plant owners
- Users who want a tool for tracking plants and building consistent care routines
- Users who prefer credible resources over inconsistent advice

3.3 Scope and Deliverables (Beta)

In Scope (Beta):

1. Verified Plant Onboarding
    a. User uploads plant photos
    b. System calls external plant ID API

    c. System returns top matches & confidence level
    d. User confirms/chooses correct identification
2. Plant Catalog
    a. Create, view, edit, and delete plant entries
    b. List view and plant details page
    c. Photo storage and verified species records
3. Care Reminders
    a. Watering reminder scheduling for each plant
    b. Basic reminder setting (Frequency & Time)
4. Extension Resources
    a. Region selection or location metadata
    b. Links to credible extension resources

3.4 Technologies / Languages

Frontend: React/Next.js + TypeScript

Backend/Database/Auth/Storage: Supabase (PostgreSQL + Auth + Storage)

External Integration: Plant Identification API (REST)

Deployment: Vercel (frontend), Supabase-hosted backend

3.5 Development Workflow

- Github will be used for version control, issues, and documentation.
- Branches: main(stable), dev(integration), and feature/* (individual works)
- Pull requests required before merges, and reviewed by other team member

- Weekly milestone versions tagged to show progress

3.6 Features

Features will be mentioned in more detail in the product backlog, but here are the requirements for features to be considered complete:

- It works from end-to-end
- It passes basic test checklist
- Documentation is updated
- It is merged through a reviewed pull request
- Secrets are kept in environmental variables (no keys committed)

Product Backlog

Prioritized list of planned features and who is responsible.

Priority	**Feature**	**Description**	**Responsible**
P0	Repo & workflow	Repo setup, branches, PR rules, issues	Travis
P0	UI foundation	Navigation + core scenes	Aran
P0	Auth	Signup/login + user sessions	Travis
P0	Photo Upload	Upload/store plant images	Travis
P0	Plant ID Integration	API call + parse results	Travis
P0	Confirm ID + save plant	UI confirm + DB write	Travis
P0	Plant catalog	List/detail/edit/delete	Aran

P0	Reminders v1	Schedules for each plant + delivery method	Travis
P1	Resources section	Region selection + extension links	Aran
P2	Dashboard	Upcoming reminders overview	Aran
P2	UX Polish	Empty states + error states, and error handling	Both

Project Plan

Key milestones, deliverables, dates

Date Range	**Milestone**	**Deliverable**
Feb 2 - Feb 7	Requirements + UI Beta Draft	Finalized beta scope + wireframes
Feb 8 - Feb 14	Backend Foundation	DB schema + auth + storage
Feb 15 - Feb 21	Onboarding	Photo Upload + create plant flow
Feb 22 - Feb 28	Plant ID Integration	API Integrated + confirmation UI
Mar 1 - Mar 7	Catalog Core	CRUD Catalog + plant detail pages
Mar 8 - Mar 14	Reminders v1	Schedule + reminder delivery approach
Mar 15 - Mar 21	Resource section	Region selection + location metadata & extension links
Mar 22 - Mar 28	Beta release	Deployed beta + demo walkthrough
Mar 29 - Apr 12	Testing + polish	Bug fixes + usability improvements

Apr 13 - Apr 20	Final packaging	Docs + final demo + submission materials

GitHub Link

https://github.com/s1vart/senior-project

References

Fogg, B. J. (2009). *A behavior model for persuasive design.* Proceedings of the 4th International Conference on Persuasive Technology. https://doi.org/10.1145/1541948.1541999

Waldchen, J., & Mader, P. (2018). *Plant species identification using computer vision techniques: A systematic literature review.* Archives of Computational Methods in Engineering, 25(2), 507-543. https://doi.org/10.1007/s11831-016-9206-z

Congressional Research Service. (2024). *The Agricultural Cooperative Extension System: An overview (R48071).* https://www.congress.gov/crs-product/R48071

Signed by Your Advisor

By signing below, the advisor confirms review and approval of the proposed project.

Advisor Name: Prof. Carrie Lapaire Harmon

Signature:

Date: